



Does conditioning leave a structural memory?

Large-area evidence for the dislocation picture

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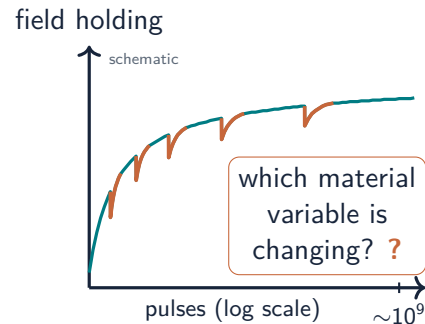
Today we ask whether conditioning leaves a structural memory in the metal. Last year at MeVArc we showed our dislocation model and STEM cross-sections of a modified near-surface layer, and we closed by asking for more comparative forensics. This talk brings the large-area answer on copper, and a first look at stainless steel. We start with what conditioning does.

Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.^a
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.^a
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.^{a,b}
- ▶ The field, more than the arcs, drives conditioning.^{a,b}

Something in the metal stores the improvement. We do not know what.^a



Conditioning works. What does it change?

Conditioning works, but we do not know what it changes in the metal. Every high-gradient device reaches its usable field by conditioning. The breakdown rate rises roughly as the thirtieth power of the field. Conditioning tracks the number of high-field pulses, not the number of breakdowns, so the field, more than the arcs, drives it. The curve is a schematic: field holding climbs over about a billion pulses. Something in the metal stores that improvement.

[optional: Walter set out these clues in his invited talk this morning.]

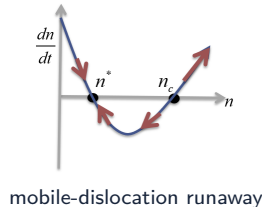
Our answer starts from a model.

Sources: ^aWuensch et al., Rev. Mod. Phys. 98, 025004 (2026); ^bDegiovanni et al., PRAB 19, 032001 (2016).

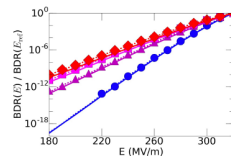
Our model: breakdown as collective dislocation dynamics



- Field holding rises FCC→BCC→HCP^d — matching the order of barriers to dislocation motion.
- The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.^{a,b}
- The same model, fitted to the $\sim E^{30}$ rate and its temperature dependence,^d then reproduced the histogram of dark-current spikes.^c



mobile-dislocation runaway



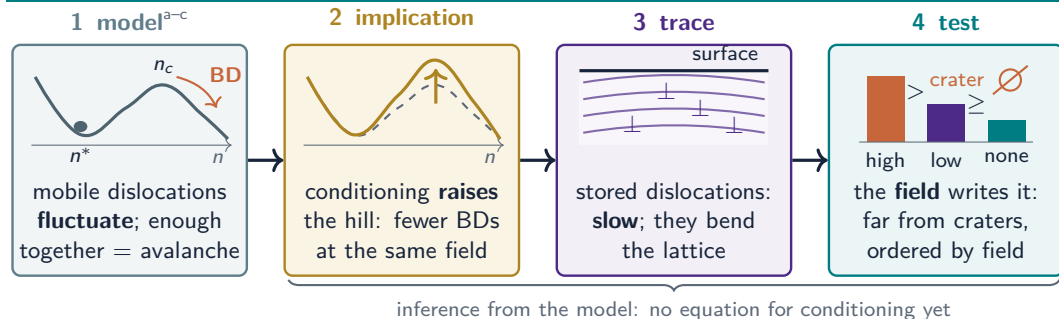
breakdown rate vs field

Our model treats breakdown as collective dislocation dynamics. Field holding rises from FCC to BCC to HCP metals, matching the order of barriers to dislocation motion. In MDDF, the mobile dislocations driven by the Maxwell stress fluctuate, and breakdown nucleates when that population runs away. We fitted the same model to the thirtieth-power rate and its temperature dependence, and it then reproduced the histogram of dark-current spikes, which was a prediction. Its central prediction for this talk is that conditioning evolves the near-surface dislocation structure. Next: what that means for a microscope.

Central prediction: conditioning *evolves the near-surface dislocation structure*.^d

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); ^dWuensch et al., RMP 98, 025004 (2026).

What the model says the microscope should see



What the model says the microscope should see

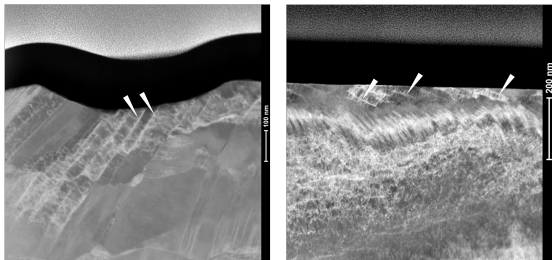
This slide turns the model into something a microscope can look for. The mobile population is fast: it fluctuates and decays back. Conditioning means fewer breakdowns at the same field, so the barrier, the hill in this picture, must rise. A memory that lasts about a billion pulses must then sit in the slow, stored dislocations that set the hill, and those bend the lattice, which EBSD reads. The field writes the memory, so we look far from craters and expect the order to follow the local field. The model predicts where and in which order; it does not predict size, depth or sign. Last year's STEM gave a first look.

- In the model, a $\sim 10^9$ -pulse memory must sit in stored, EBSD-visible dislocations.
- The field writes it: look far from craters; expect the order to follow the local field.

predicts where, and in which order **does not predict** size, depth, or sign

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); Wuensch et al., RMP 98, 025004 (2026).

Identified using STEM at specific locations



Cross-sectional STEM of conditioned hard (as-machined) Cu: dislocation-poor zone in the top ~ 200 nm.^a
STEM: HUJI, I. Popov

- ▶ **Direct** imaging of an altered near-surface dislocation structure in conditioned regions.^a
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.^a
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.^b
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

Seen at specific locations. The large-area view was missing.

Sources: ^aJacewicz et al., JAP 137, 193302 (2025); ^bKorsbäck et al., PRAB 23, 033102 (2020).

Identified using STEM at specific locations

STEM has already imaged an altered near-surface dislocation structure, but only at specific locations. These are our 2025 cross-sections of conditioned hard copper, imaged at HUJI by Inna Popov. Under the protective coating, the top 200 nanometers or so of the field-exposed region is dislocation-poor, with fewer walls. We see fewer walls at 300 and at 30 kelvin, and the denuded layer is clearest at 30 kelvin. This fits an independent clue: soft copper conditions far more slowly than hard copper, so the starting population matters. Two cautions: FIB preparation plants its own dislocations, so we trust only large-scale structure, and one cross-section cannot speak for the whole electrode. The large-area view was missing. But first: can a stress this small write the structure at all?

[if asked: the paper gives no thickness; about 200 nanometers is read from its Figure 10. Hard copper conditioned within about 25 million pulses; soft copper needed some 600 to 900 million.]

A sub-yield stress, repeated a billion times

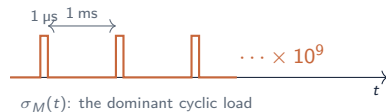


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: **0.028 MPa^c**

$\sim 10^3 \times$ below the yield of annealed Cu^c



- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.^{a,b}
- ▶ Here: $\sim 10^9$ conditioning pulses.^c
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the dominant cyclic load.^c
- ▶ MDDF needs only the stress that moves **mobile dislocations in a frustrated crystal**: far below yield.

**The model predicts a large-area signature.
Is it there?**

The load is tiny, but it is repeated about a billion times. The Maxwell stress at 80 megavolts per meter is 0.028 megapascals, about a thousand times below the yield of annealed copper. Fatigue at ten to the eight to ten to the ten cycles rearranges dislocations irreversibly well below yield, which is our regime. Pulsed DC means no RF pulsed heating and no bulk currents, so the Maxwell stress is the dominant cyclic load. MDDF needs only the stress that moves mobile dislocations in a frustrated crystal, far below yield: a dense, pinned network in which weak internal stresses, not the applied one, decide where mobile segments move.

[optional: Jianyu, with Guodong Meng, and Veronika show field-driven deformation at the nanoscale this afternoon, at 16:00 and 16:30.]

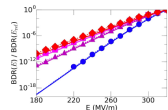
So the model predicts a large-area signature: is it there?

Sources: ^aMughrabi, MMTB 40, 431 (2009); ^bStanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007); ^cAshkenazy et al., arXiv:2606.19192 (2026).

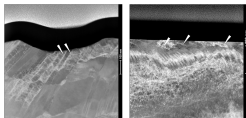
Last year we asked for more forensics. Here it is.



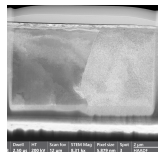
MeVArc 2025



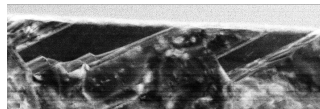
MDDF model



local STEM denuded zone^a



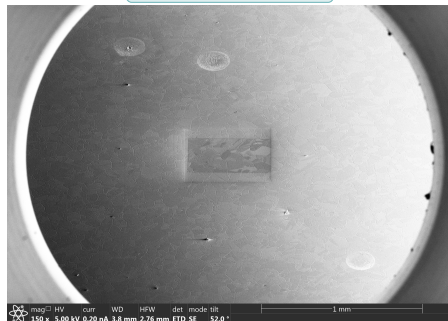
FIB lamella,
HAADF overview



cross-section STEM
at the surface

STEM: HUJI, I. Popov

MeVArc 2026



Large-area EBSD across one cathode

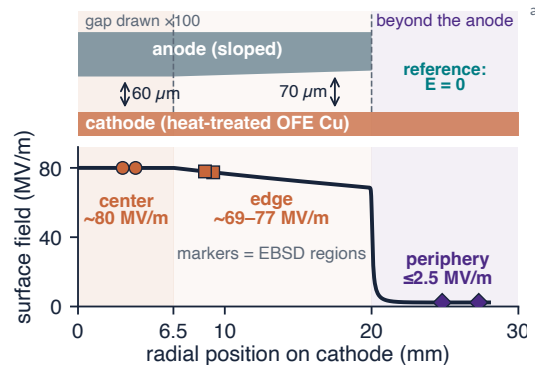
Last year we asked for more forensics. Here it is.

Last year we asked for more comparative forensics, and this slide shows what came back. On the left is what we showed in 2025: the model, and local STEM of the denuded zone, from a FIB lamella overview to a cross-section at the surface. We closed with two questions: other materials, and more samples. On the right is the large-area answer, an EBSD survey across one conditioned cathode. Let us look at how that cathode was exposed.

Closed with: **other materials?** **more samples?**

Source: ^aJacewicz et al., JAP 137, 193302 (2025).

One cathode turns field exposure into a controlled coordinate



● high field (center, edge) ● periphery (≤ 2.5 MV/m) ● unexposed reference

The field gradient gives us internal controls on a single specimen.

Sources: ^ageometry: Bjelland et al., arXiv:2606.21259 (2026); Ashkenazy et al., arXiv:2606.19192 (2026), $E(r)$: Fig. 3.

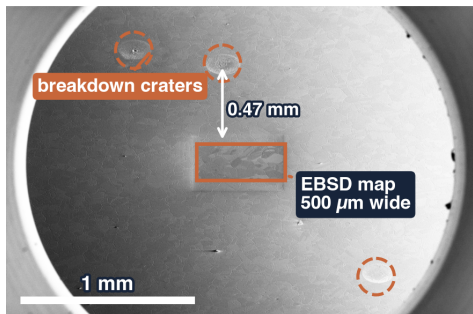
One cathode turns field exposure into a controlled coordinate

One cathode gives us a controlled range of field exposure. The anode is sloped, so radius maps onto local field: about 80 megavolts per meter under the flat center, about 69 to 77 at the edge, and below about 2.5 beyond the anode radius. The cathode is heat-treated OFE copper, conditioned with 1-microsecond pulses at 1 kilohertz, about ten to the nine pulses. We mapped nine EBSD regions: center, edge, periphery, and an identically treated reference that never saw field. Copper for high field, purple for the periphery and teal for the reference stay the same on every data slide. Comparing regions on one specimen removes cathode-to-cathode differences. [optional: this is the CERN pulsed-DC system; Victoria gave its update this morning.]

Here is what EBSD sees on that surface.

- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ $1 \mu\text{s}$ pulses at 1 kHz, $\sim 10^9$ pulses
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

EBSD maps plastic activity far from breakdown craters



- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only)^a; information depth ~ 40 nm at 15 kV^{b,c}
- ▶ $\sim 24,000$ indexed points per ROI^a
- ▶ **LAM** = local average misorientation: lattice tilt between a point and its neighbors (KAM, LOS: cross-checks)^a
- ▶ Misorientation is a proxy for GND density^d

Target: the cumulative response to the field, measured far from craters.^a

Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^b ~ 40 nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^cmeasured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011); ^dPantleon, Scr. Mater. 58, 994 (2008).

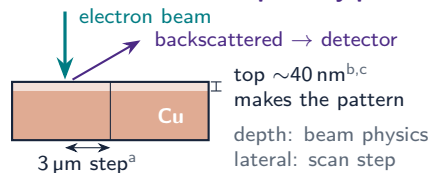
We use EBSD because it covers a large area far from breakdown craters. The copper box is one 500-micrometer map; the dashed circles are craters, the nearest about half a millimeter away. The 3-micrometer step sets only the lateral resolution; the signal comes from the top 40 nanometers of copper at 15 kilovolts, so this is a near-surface measurement. Each region has about 24,000 indexed points. Our main metric is LAM, the local average misorientation, with KAM and LOS as cross-checks, and misorientation is a proxy for the density of geometrically necessary dislocations. Our target is the cumulative response to the field, not crater damage. Next: what one pixel measures.

[if asked: a shallow FIB clean removes the oxide, under 10 nanometers by AFM, with the same protocol on all nine regions.]

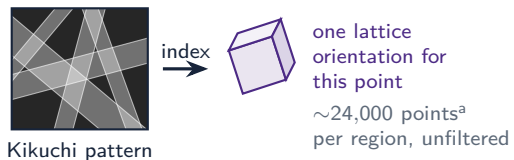
What one EBSD pixel measures



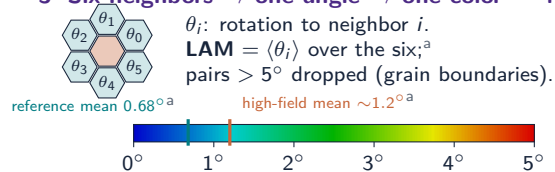
1 15 kV electron beam, point by point



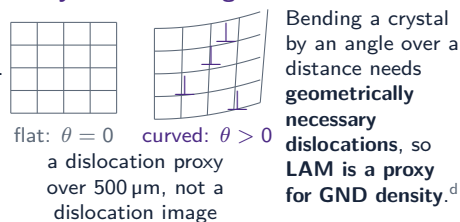
2 Pattern → lattice orientation



3 Six neighbors → one angle → one color



4 Why a rotation angle counts dislocations

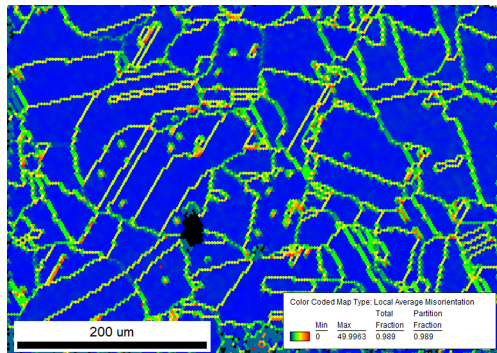


Each 3 μ m point yields one angle, read from the top ~ 40 nm of copper.^{b,c}

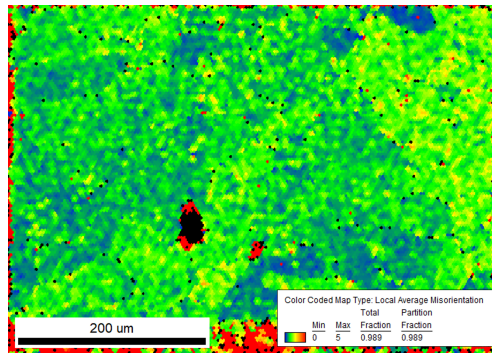
Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^b ~ 40 nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^cmeasured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011); ^dPantleon, Scr. Mater. 58, 994 (2008).

Every colored pixel on the next slides is one angle, read from the top 40 nanometers. The 15-kilovolt beam visits the surface point by point on a 3-micrometer grid. Each point gives a Kikuchi pattern, and indexing turns it into one lattice orientation. LAM averages the rotation to the six neighbors, dropping pairs above five degrees, so grain boundaries are excluded. On this scale the reference sits at 0.68 degrees and high field at about 1.2. A bent lattice needs geometrically necessary dislocations, so LAM is a dislocation proxy over half a millimeter, not a dislocation image. Next: how to read a LAM map.

How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate



0–5° scale: intragrain curvature

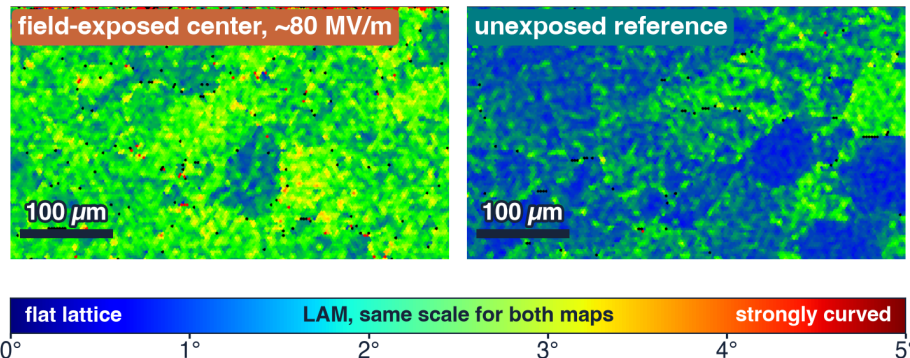
Rescaling to 0–5° exposes the intragrain misorientation we care about.

How to read a LAM map: wide-angle vs low-angle

The color scale decides what a LAM map shows. On a zero-to-50-degree scale, only grain boundaries stand out. Rescaling to zero to five degrees reveals the curvature inside the grains, the low-angle misorientation that tracks geometrically necessary dislocations. That intragrain signal is what we care about. Every map from here on uses the same zero-to-five-degree scale. Now we compare high field with the reference.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Same area, two color scales.

High-field copper contains more intragrain curvature

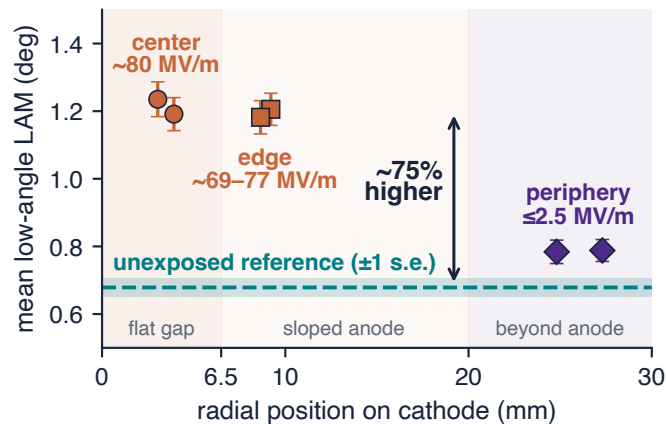


High-field copper carries more curvature inside its grains. Both maps use the same color scale and the same magnification, cropped to the same field of view. Look at the texture, not only the average color: the field-exposed grains are full of green and yellow, while the reference stays mostly blue. This pair is representative; the numbers come from all nine regions. Here are those numbers.

Same scale, same magnification: field-exposed grains carry more curvature.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Both maps cropped to the same $485 \times 279 \mu\text{m}$ field; LAM $0\text{--}5^\circ$.

The signal separates into three exposure tiers



- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The signal separates into three exposure tiers

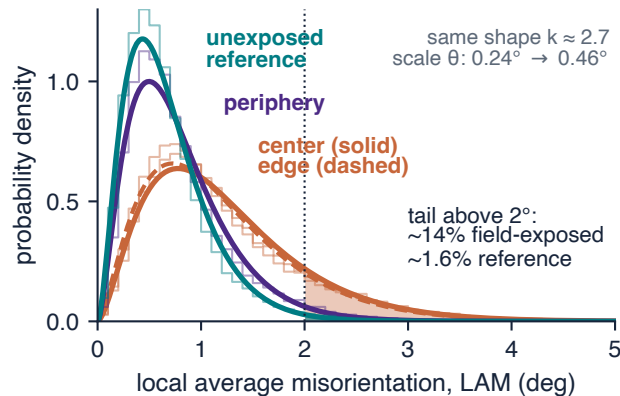
The signal separates into three tiers, and the tiers follow field exposure. Center and edge, both at high field, sit at about 1.2 degrees, and the external reference at about 0.68, so high field is about 75 percent higher. The periphery, on the same cathode, saw at most 2.5 megavolts per meter and sits in between. It shared machining, heat treatment, mounting, vacuum and handling with the high-field regions, so it is a low-field internal control. Grain size does not follow this hierarchy; the ordering follows field history. The 75 percent describes this cathode and its history, not a universal constant. Next, the shape of the distributions.

[if asked: breakdown density falls from about 24 to 12 to 5 per square centimeter from center outward; center and edge differ twofold in breakdown density but not in LAM.]

The ordering follows field history; grain size does not track it.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Error bars: grain-corrected standard errors; band: unexposed reference ± 1 s.e.

Field exposure builds a high-misorientation tail



LAM tail above 2°

~ 0.14 field-exposed
 ~ 0.016 reference

$\sim 8\times$ **increase**

Gamma shape $k \approx 2.7$ fixed; the scale roughly doubles.

Field exposure builds a high-misorientation tail

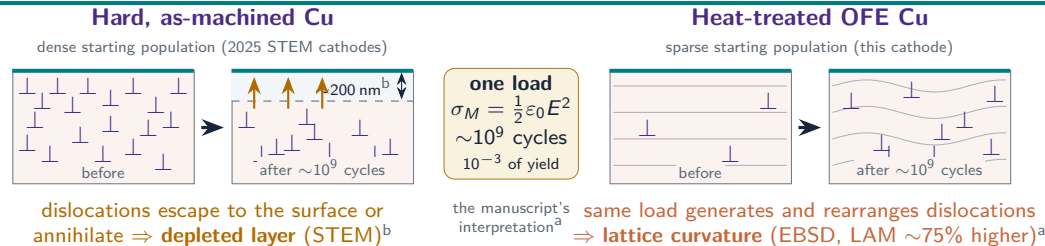
Field exposure does more than shift the distribution: it builds a high-misorientation tail. The dotted line marks two degrees, and the shaded areas are the tails beyond it. The fraction above two degrees is about 0.14 at high field against 0.016 on the reference, roughly eightfold. A gamma fit keeps its shape, k near 2.7, while the scale roughly doubles. The whole distribution stretches, and a uniform offset cannot do that. LAM, KAM and LOS all agree. So what does this say about the model?

[if asked: KS, chi-squared and Anderson–Darling tests all separate the distributions; we quote effect sizes and grain-corrected errors rather than pixel-level p-values.]

The whole distribution stretches; a uniform offset cannot do that.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). One ROI per tier (histograms faded, gamma fits solid, edge dashed); shaded: tail above 2° .

The large-area test supports the model's prediction



- To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts.^a
- The measured order follows the simulated local field:
center $\approx$ edge $>$ periphery $>$ reference.^a

Before, dislocations were plausible.
Now a dislocation-sensitive state tracks field exposure.

Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026), discussion; ^bJacewicz et al., JAP 137, 193302 (2025).

The large-area test supports the model's prediction. To our knowledge, this is the first large-area observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts and following the simulated local field. The cartoon is our interpretation from the manuscript: one load acting on two starting populations. Dense, as-machined copper loses dislocations near the surface, the layer STEM saw; heat-treated copper has little to lose, so the same load builds the curvature EBSD records. Size, depth and sign were not predicted. Which of the stored dislocations matter? Those near enough to the surface to exchange stress with it, and that is the next slide.

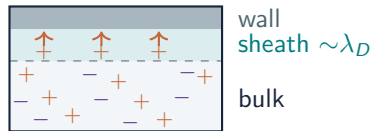
[if asked, the theory–experiment record: predicted observable, a change in the stored near-surface population; measured, LAM; mapping, a relative GND proxy under one protocol; no parameters fitted to the EBSD data; no prediction of the size of the change.]

[if asked about E_S : its simulated profile has the same order; a candidate basis, not a measurement.]

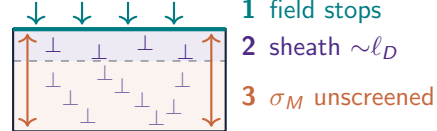
Elastic screening: a sheath of dislocations at the surface



Plasma at an absorbing wall schematic



Copper under the Maxwell traction schematic



1 $\lesssim 0.1 \text{ nm}$
field penetration
(Thomas–Fermi)

2 $\ell_D \sim 25 \text{ nm}^a$
screening of weak internal
fluctuations;^{b,c} only this layer
talks to the surface

3 $\sim \text{mm}$
applied load, not screened
(dislocation stress is
self-equilibrated)^d

In this picture, weak screened internal stresses set the structure; the strong unscreened load drives the mobile dislocations through it.

Sources: ^aGroma et al., PRL 96, 165503 (2006); ^bLemaître et al., PRE 104, 024904 (2021); ^cLivne et al., PRE 107, 055004 (2023); ^dAshkenazy et al., arXiv:2606.19192 (2026).

Elastic screening: a sheath of dislocations at the surface

Three length scales decide what the surface sees; in the plasma on the left, only a sheath one Debye length thick reaches the wall. First, the field stops within 0.1 nanometers. Second, weak internal fluctuations are screened within about 25 nanometers, and only that layer, which EBSD samples, talks to the surface. Third, the applied load is not screened, since dislocation stress is self-equilibrated; it reaches millimeters. Copper is a plasma of dipoles, Debye-like for fluctuations and dielectric-like for the mean load: it reduces, it does not nullify; and the surface is an absorbing wall: a sheath of one screening length sets what it sees, hence emission and breakdown. In our picture, weak screened internal stresses set the structure, and the strong unscreened load drives the mobile dislocations through it. Next: what copper leaves open, and where we go.

[two time constants: think of the mobile dislocations as the electrons and the stored network as the ions. The one-microsecond pulse is far above the network's plasma frequency, so it passes unscreened; the mobile population responds within the pulse and screens the slow internal fluctuations. Ten-to-the-nine repetitions act like a ponderomotive force—a slow, cumulative, nonlinear push of a fast drive—and the Maxwell stress itself is the ponderomotive force of the field on the metal.] [if asked about the dielectric analogy of earlier versions: dislocations are dipoles, not free monopoles, so the mean load is reduced, never canceled.]

[if asked for a real plasma example: a 13.56 MHz processing plasma. The drive lies between the ion and electron plasma frequencies (about 3 MHz and 1 GHz at 10^{10} per cubic centimeter in argon): electrons screen within each cycle and form the sheath, ions see only the time-averaged sheath potential, and in an asymmetric discharge the fast drive even produces a steady self-bias. Here the mobile dislocations are the electrons, the stored network the ions, the pulse train the RF, and the conditioning memory the time-averaged, rectified effect.]

[if asked why conditioning does not need to screen the applied stress: it cannot, and it need not. What changes is the medium's response function—stored density, sources, ℓ_D —i.e. the threshold of the mobile-population instability, not the drive.] [if asked: the 25 nanometers is an order-of-magnitude estimate, Debye-like screening in a two-dimensional single-slip dislocation gas (Groma 2006).]

Copper shows the effect, not universality



What is shown

- ▶ Large-area structural difference
- ▶ Three metrics agree (LAM, KAM, LOS)
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

What remains open

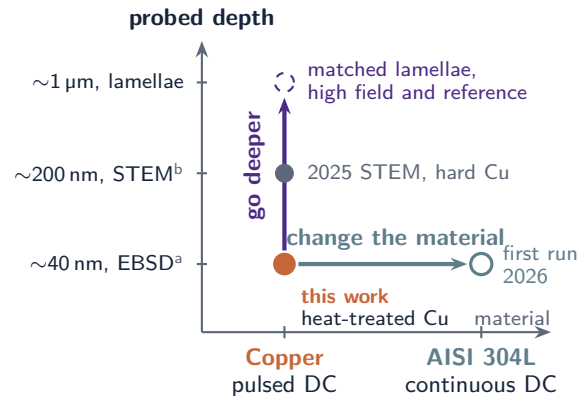
- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Limits of this dataset: field level and breakdown density vary together by construction ($\sim 24, 12, 5$ breakdowns per cm^2 , center outward); LAM values are relative, compared only between regions mapped under the same step and protocol.

A stronger test changes both the material and the loading history.

Copper shows the effect; it does not show universality. What is shown: a large-area structural difference, three metrics that agree, internal and external controls, and support for the MDDF prediction. What remains open: one conditioned cathode in one experiment, no full depth profile, no grain-orientation dependence yet, no further cathodes and no second material. On this cathode, field level and breakdown density vary together, and LAM is a relative proxy, compared under one protocol. A stronger test changes the depth, the material, and the loading history: depth and material are the next slide; the loading history comes with the steel.

Next: go deeper, and change the material



- ▶ Cross-sections from a mapped high-field region and from the reference, through the top $\sim 1 \mu\text{m}$: do the EBSD layer ($\sim 40 \text{ nm}$) and the STEM denuded zone ($\sim 200 \text{ nm}$) belong to one depth profile — is the denuded zone the sheath ($\approx 8 \ell_D$)?
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ Stainless steel under continuous DC, with Consorzio RFX — next slide

Next: go deeper, and change the material

Our next steps run along two axes, depth and material. Today’s result sits at about 40 nanometers in heat-treated copper; the 2025 STEM depletion in hard copper was about 200 nanometers deep, read from the scale bar of the paper’s Figure 10. Going deeper means cross-sections from a mapped high-field region and from the reference, through the top micrometer. They ask whether the EBSD layer and the denuded zone belong to one depth profile, and whether the denuded zone is the sheath, about eight screening lengths wide. We carry over the artifact controls from the 2025 red herrings. Depth continues our own STEM line, published with Uppsala. Along the material axis comes stainless steel under continuous DC, with Consorzio RFX, on the next slide.

Depth continues our STEM line; material tests the mechanism itself.

Sources: ^aCASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^bJacewicz et al., JAP 137, 193302 (2025).

Steel: different material, different structural evolution



What differs from copper

- ▶ Lower stacking-fault energy: planar slip, different dislocation mobility
- ▶ Continuous DC: a steady traction of ~ 16 kPa, not $\sim 10^9$ pulses
- ▶ Metastable austenite is a built-in strain gauge: ϵ and α' martensite mark plastic strain above a few 0.1 %^a

The dislocation picture expects a different near-surface evolution. The test: does any field-correlated change appear at all?



Consorzio RFX — De Lorenzi, Pilan, Spada

First run with Consorzio RFX

- ▶ AISI 304L at the RFX HVPTF: continuous DC, 90 h and 73 h at 59–61 MV/m; zero breakdowns, stopped at emission switch-on
- ▶ 26 EBSD maps (HUJI, I. Popov): the exposed apex lies inside or below the unexposed range
- ▶ References exist (unexposed electrodes, side regions) but no same-area before-and-after; with 2–4 maps per group a field effect cannot yet be resolved

RFX HV program: N. Pilan, next talk (12:30).

Copper supports the memory; steel asks whether it is universal.

Source: ^aBayerlein et al., MSEA 114, L11 (1989).

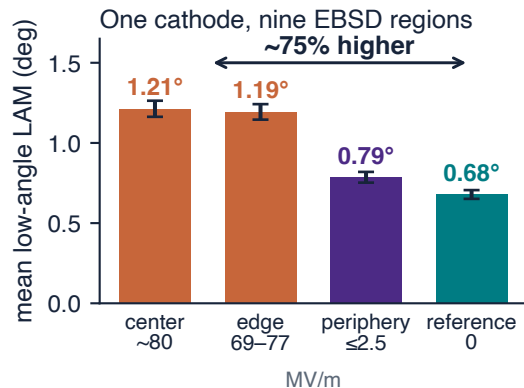
Steel is a different material, so we expect different structural evolution. Lower stacking-fault energy means planar slip, and continuous DC gives a steady traction of about 16 kilopascals not a billion pulses. Its metastable austenite is a built-in strain gauge: epsilon and alpha-prime martensite mark plastic strain above a few tenths of a percent. With Consorzio RFX, Antonio De Lorenzi, Nicola Pilan and Emanuele Spada, two 304L electrodes ran 90 and 73 hours at 59 to 61 megavolts per meter, zero breakdowns, stopped at emission switch-on. On 26 EBSD maps the exposed apex lies inside or below the unexposed range. References exist, but there is no same-area before-and-after. With two to four maps per group, a field effect cannot yet be resolved. Copper supports the memory; steel asks whether it is universal. Nicola will tell you about the RFX program next.

[if asked: we have not yet looked for martensite on an exposed surface. Next on steel: one electrode mapped before and after being driven into breakdowns, with epsilon and alpha-prime in the phase list, so that absent martensite becomes a strain bound. 80 percent power at one standard deviation needs about 17 maps per group.]

Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** ($\sim 75\%$ higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Conditioning appears to leave a subsurface memory. Our model predicts that conditioning evolves the near-surface dislocation structure. STEM saw it locally in 2025, and large-area EBSD now supports it. The four bars are the tier means: 1.21 and 1.19 degrees at center and edge, 0.79 at the periphery and 0.68 on the reference, about 75 percent higher at high field. Depth and stainless steel now test whether the memory survives a change of scale, alloy and loading. The paper is on arXiv, 2606.19192. To close, what we found and what we are asking for.

Last year: “other materials? more samples?” — here is the large-area answer.

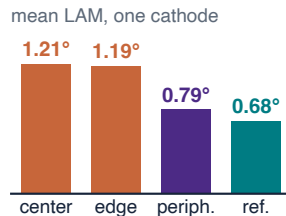
Copper result: Ashkenazy et al., arXiv:2606.19192 (2026).

Conclusions, and an invitation



1. Conditioning leaves a structural memory in the near-surface dislocation population — supported by single cases (one large-area cathode, three STEM conditions), not yet replicated.^a
2. Our picture: it lives within a screening length of the surface — the layer EBSD sees, and the layer that sets emission.
3. What would settle it: the same area mapped before and after; a depth profile; a second material (steel, with Consorzio RFX).

Talk to me: new materials, other electrode configurations, cross-facility comparisons, same-area before-and-after mapping — let us test this together.



arxiv.org/pdf/2606.19192

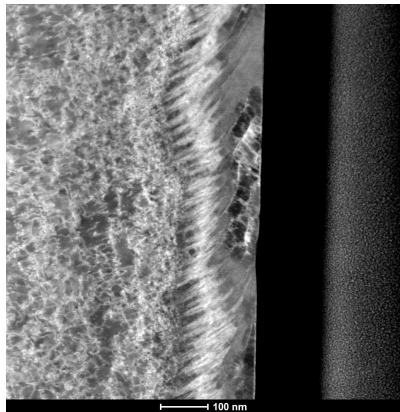
^a Ashkenazy et al., arXiv:2606.19192

Three conclusions, and an invitation. Conditioning leaves a structural memory in the near-surface dislocation population; it is supported by a handful of single cases, one sample per condition, and not yet replicated. In our picture it lives within a screening length of the surface, the layer EBSD sees and the layer that sets emission. What would settle it is the same area mapped before and after, a depth profile, and a second material, steel with Consorzio RFX. The take-away. Breakdown is collective dislocation dynamics, so conditioning must leave a structural trace, and we have now read that trace over a large area: the same cathode, ordered by field, with the conditioned regions carrying more near-surface misorientation. If you take one picture home: the field stops at the surface, the load does not, and in between sits a sheath of dislocations about a screening length thick. The sub-yield load, a billion times over, slowly rearranges that sheath, and the sheath decides what the surface emits. Conditioning is the memory of that rearrangement. The large-area map rests on one cathode; with the STEM cases, hard copper at 300 and at 30 kelvin and soft copper, we have a handful of singular cases, one sample per condition. The next word belongs to a before-and-after map and to a second material. Supported on copper, open on steel. So talk to me about new materials, other electrode configurations and same-area mapping, and the QR code opens the paper.

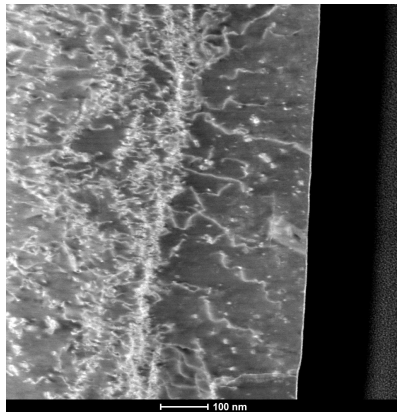
[optional: Mircea's resistivity measurements, at 11:30 just before us, probe the same state.]

Thank you. Nicola will tell you about the RFX program next.

Backup: field-exposed vs reference STEM (2025)



Field-exposed, 300 K: fewer walls



Reference, 300 K: walls to surface

STEM: HUJI, I. Popov.

Source: Jacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.

Backup: field-exposed vs reference STEM (2025)

This is the field-exposed versus reference pair from the 300-kelvin cathode. On the field-exposed side the dislocation walls are strongly reduced, while the reference keeps walls up to the surface. In the published paper this pair is Figure 9; the zone the paper calls denuded, and shows most clearly, is on the 30-kelvin cathode, Figure 10. The paper states no thickness. Back to the questions.

[if asked: the 200 nanometers is read from the Figure 10 scale bar.]